

Methodology

Section 3 cast the wave release coordination problem as a simulation-based decision problem with a combinatorial decision space and an objective $C_{max}(W; M)$ realized by an event-driven simulator. Direct enumeration is intractable, so this section develops two analytical tools that operate on the structured representation $\Phi(W)$ of Section 3 and on a finite partition of Φ -space. The first, a Bound-and-Gap framework (Section 4.1), decomposes the value of wave-structure information into two non-negative components and rests on a decomposition theorem together with an equivalent to Smart-Predict-then-Optimize regret on a partition-constant predictor class. The second, a Model-Dominance Hedge Rule (Section 4.2), reduces the dispatch decision under elevator-model uncertainty to a closed-form policy and rests on a minimax-collapse theorem together with an equivalence to a Wasserstein distributionally robust solution under chain dominance. Each tool is stated here as a theorem with a proof sketch; Section 4.3 lists the structural predictions that Section 5 tests.

4.1 The Bound-and-Gap framework

The Bound-and-Gap framework treats wave-structure information, the knowledge that a wave's Φ -coordinates lie in a particular region of Φ -space, as an information-theoretic object and asks how much of the value of acting on that information is recoverable by a feature-based policy and how much is structurally unrecoverable. We construct the framework in three steps. Section 4.1.1 establishes the decomposition $GAP = H_{up} + M_\Phi$ as Proposition 1. Section 4.1.2 gives the policy component M_Φ a second reading as a Smart-Predict-then-Optimize regret on the partition-constant predictor class (Theorem 1). Section 4.1.3 turns the pair (H_{up}, M_Φ) into a diagnostic reading and shows how each component responds to refinement of the partition.

4.1.1 Decomposing the value of wave-structure information

Fix a (regime, model, size) cell. Partition Φ -space by a finite scheme Q ; the working instance throughout the paper is the 2×2 quartile partition of the (C, I) plane into four corners $Q = \{HC \cdot HI, HC \cdot LI, LC \cdot HI, LC \cdot LI\}$. Let m_q denote the corner-conditional median make-span of waves drawn from corner $q \in Q$, and let m_0 denote the median make-span of a wave drawn at random from the unpartitioned pool. Two extreme corners, $q_{max} = \operatorname{argmax}_{q \in Q} m_q$ and $q_{min} = \operatorname{argmin}_{q \in Q} m_q$, bound what the partition can in principle reveal; a third corner, q_Φ , is the corner that a Φ -informed policy selects from the sign pattern of an ordinary-least-squares fit on the cell. Define the oracle upper bound, the Φ -informed lower bound, and their gap:

$UB = \frac{m_{q_{max}} - m_{q_{min}}}{m_0}, \quad LB = \frac{m_0 - m_{q_\Phi}}{m_0}, \quad GAP = UB - LB$	
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Proposition 1 (Bound-and-Gap decomposition). For any cell and any finite partition Q ,

$GAP = H_{up} + M_\Phi, \quad H_{up} = \frac{m_{q_{max}} - m_0}{m_0}, \quad M_\Phi = \frac{m_{q_\Phi} - m_{q_{min}}}{m_0}$	
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Proof. Substituting the definitions in (4.1),

$UB - LB = \frac{m_{q_{max}} - m_{q_{min}}}{m_0} - \frac{m_0 - m_{q_\Phi}}{m_0} = \frac{m_{q_{max}} - m_0}{m_0} + \frac{m_{q_\Phi} - m_{q_{min}}}{m_0} = H_{up} + M_\Phi$	
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Because $m_{q_{max}} \geq m_0$ and $m_{q_\Phi} \geq m_{q_{min}}$ by construction, both components are non-negative, and $GAP \geq 0$ strictly whenever the partition is non-degenerate. The two components admit distinct

interpretations. H_{up} is the partition-intrinsic upper-tail headroom, the relative make-span penalty of the worst corner against the unpartitioned median, present even when Φ selects perfectly and not addressable by improving the policy. M_Φ is the Φ -policy miss, the relative cost of Φ selecting q_Φ rather than the oracle-best corner q_{min} , a residual that better feature engineering could in principle recover. Section 5 reports the empirical magnitudes of both components across the publication-scale grid.

4.1.2 The policy component is an SPO regret

The policy component M_Φ admits a second reading that situates the Bound-and-Gap framework inside the prediction-to-decision regret literature. Treat which corner to release a wave from as a decision over Q ; the true per-corner cost vector is $c = (m_q)_{q \in Q}$, and the oracle decision is

$z^*(c) = \arg \min_{q \in Q} m_q$	
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Following Elmachetoub & Grigas (2020), the Smart-Predict-then-Optimize (SPO) loss of any predictor $\hat{c} \in \mathbb{R}^{|Q|}$ against the true cost c is the realized-minus-oracle cost of the corner it induces:

$l_{SPO}(\hat{c}, c) = c[z^*(\hat{c})] - c[z^*(c)] = m_{z^*(\hat{c})} - m_{q_{min}}$	4.4
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A partition-constant predictor is a cost vector that is constant on each corner, the natural predictor class on a finite partition. The Φ -informed predictor, denoted $\hat{c}_\Phi$, is the partition-constant predictor whose induced decision is q_Φ .

Theorem 1 (SPO-regret equivalence). The policy component M_Φ of the Bound-and-Gap decomposition equals the SPO regret of the Φ -induced partition-constant predictor, normalized by m_0 :

$M_\Phi = \frac{m_{q_\Phi} - m_{q_{min}}}{m_0} = \frac{l_{SPO}(\hat{c}_\Phi, c)}{m_0}$	
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Proof sketch. The Φ -informed predictor induces the decision q_Φ , and the oracle induces q_{min} by definition. Substituting into (4.4) gives $l_{SPO}(\hat{c}_\Phi, c) = m_{q_\Phi} - m_{q_{min}}$, and dividing by m_0 reproduces M_Φ in (4.2). The equivalence is structural rather than asymptotic; it holds at every cell of the partition.

Theorem 1 positions the Bound-and-Gap framework as the partition perspective on SPO: the prediction-to-decision regret literature bounds an algorithm's loss at training time over a hypothesis class, where M_Φ is a post-hoc information-value gap on a fixed partition. The cell-median is used throughout for consistency with this equivalence, which is stated for the median; the cell-mean version yields a parallel result. The contribution of the Bound-and-Gap framework is therefore not the SPO regret itself, which is a known object, but the decomposition that separates the SPO regret M_Φ from an independent partition-intrinsic headroom H_{up} .

4.1.3 Reading the decomposition

Because H_{up} and M_Φ are non-negative and independently interpretable, their pair gives a structural reading of where a cell's wave-design value sits. Table 1 maps the four regions of (H_{up}, M_Φ) to four operational interpretations.

Table 1. Reading the Bound-and-Gap decomposition

H_{up}	M_Φ	Interpretation
Large	Small	Φ is already near-optimal; the residual is structural, and elevator-side capacity is the lever

Small	Large	Corners are flat and Φ is miscalibrated; feature engineering is the lever
Large	Large	The partition contrast is high and Φ misses it; feature expansion is the priority
Small	Small	The partition gives little leverage; wave design is not the lever here

The two components also respond differently to refinement of the partition.

Corollary 1 (refinement monotonicity). If a partition Q' refines Q , then $H_{up}(Q') \geq H_{up}(Q)$ — refinement of the partition can only raise the worst-corner median against the unpartitioned baseline. The policy component M_Φ carries no such guarantee, because q_Φ is itself a partition-dependent choice.

The two readings in Table 1 are of components, not of which operating regime a given dispatch policy will favor; Section 5 shows the realized value of structured wave-and-dispatch design to be broad across regimes rather than sharply regime-specific.

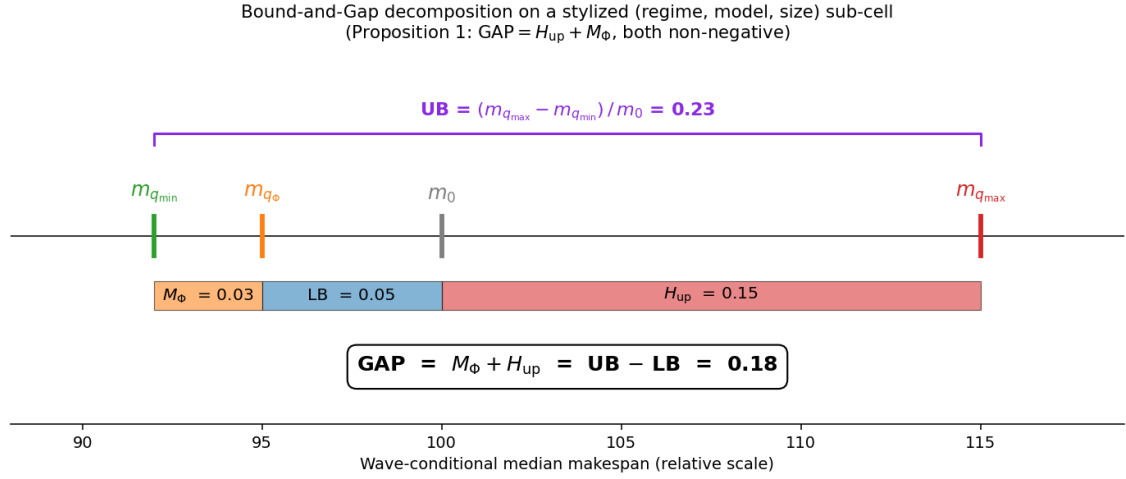


Fig. 1 The Bound-and-Gap decomposition on a stylized (regime, model, size) sub-cell. The four corner medians $m_{q_{min}}, m_{q_\Phi}, m_0, m_{q_{max}}$ tile the oracle spread UB into three contiguous segments (M_Φ, LB, H_{up}) , along a relative make-span number line. Proposition 1 reads off the equality $GAP = H_{up} + M_\Phi = UB - LB$; both M_Φ and H_{up} are non-negative by construction.

4.2 The Model-Dominance Hedge Rule

The Model-Dominance Hedge Rule addresses uncertainty over which member of the elevator-model family $\mathcal{M} = \{M_1, M_2, M_3\}$ describes the warehouse's elevator subsystem at decision time. Section 3 Assumption A5 lists this uncertainty as exogenous to the wave-composition decision; the rule below provides a closed-form policy together with a worst-case loss certificate against any member of $\mathcal{M}$. The construction proceeds in three steps: a minimax setup (4.2.1), a collapse theorem under chain dominance with an equivalent Wasserstein-DRO reading (4.2.2), and a perturbation extension to approximate dominance (4.2.3).

4.2.1 Model uncertainty and the minimax corner

The elevator model $M \in \mathcal{M}$ is exogenous to wave composition (Assumption A5), and the

operator is not assumed to identify M online. The natural conservative response is the minimax wave-corner selection

$c^* = \arg \min_{c \in Q} \max_{M \in \mathcal{M}} s[C_{\max}(W; M) W \in c]$	
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where s is any monotone summary statistic of the make-span distribution, which median, mean, or any quantile. Evaluated naively, c^* requires the operator to compute s under every $M \in \mathcal{M}$ and select against the worst. The Hedge Rule removes that requirement by exploiting a structural relation among the models in $\mathcal{M}$.

The load-bearing condition is chain dominance: for every wave W ,

$C_{\max}(W; M_1) \leq C_{\max}(W; M_2), \text{ almost surely,}$	
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with M_3 the stochastic extension of M_2 obtained by adding lognormal noise of scale σ_{M_3} to the timing primitives (3.7). In other words, throughput aggregation never overstates make-span relative to true co-occupancy batching, because aggregating per-AMR throughput absorbs the trip-time penalty that explicit co-occupancy charges per elevator trip. Chain dominance is an empirical claim about the model family $\mathcal{M}$; Section 5 verifies it on matched waves at publication scale (99.2% average, 96.0% worst cell).

4.2.2 Collapse under chain dominance and its DRO equivalence

Theorem 2 (minimax collapse and Wasserstein-DRO equivalence). Under chain dominance (4.7), the minimax corner selection (4.6) collapses to the corner optimal for the dominant model,

$c^* = \arg \min_{c \in Q} s[C_{\max}(W; M_2) W \in c]$	
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and this corner coincides with the solution of the Wasserstein-1 distributionally robust problem with ambiguity ball $B_{\rho_c}(M_1) = \{Q: W_1(M_1, Q) \leq \rho_c\}$ of corner-calibrated radius $\rho_c = W_1(M_1, M_2 | c)$, centred at the nominal model M_1 :

$\arg \min_{c \in Q}$	
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